

SALES PERFORMANCE ANALYSIS DASHBOARD

PROJECT DOCUMENTATION

Submitted By	Sharulatha B
Course	AI-driven Data Analytics
Tool Used	Microsoft Power BI
Date	June 28, 2026

PROJECT OVERVIEW

This project leverages Power BI to analyze and visualize sales data across multiple categories — Furniture, Clothing, and Electronics — using datasets of orders, detailed transactions, and monthly sales targets. The primary objective is to apply DAX (Data Analysis Expressions) for creating calculated measures such as total sales, profit margins, and category-wise performance, while comparing actual results against predefined targets.

OBJECTIVES OF THE PROJECT

The objective of this project is to analyze and visualize sales performance using Microsoft Power BI. It aims to demonstrate how data from multiple sources — List of Orders, Order Details, and Sales Targets — can be integrated, cleaned, and modeled to generate meaningful insights.

Target Audience

- Executive Leadership → Strategic overview of performance
- Sales Operations → Category and region-wise analysis
- Finance Teams → Profitability and margin tracking

DATA SOURCES & ARCHITECTURE

Source Systems	Local Excel/CSV files (List of Orders, Order Details, Sales Target)
Data Volume	500 rows covering Jan–Nov 2026

Storage Mode	Import mode for optimized performance
--------------	---------------------------------------

DATA PREPARATION & TRANSFORMATION (ETL)

Tool Used: Power Query Editor

Key Cleanups

- Merged Orders and Order Details using Order ID
- Converted text dates into proper Date type
- Removed duplicates and null values
- Applied TRIM/CLEAN for text fields
- Unpivoted monthly target columns for normalized comparison

Custom Functions

- Profit Margin calculation: $(\text{Profit} \div \text{Sales}) \times 100$
- Automated Target Achievement logic
- Dynamic Date formatting

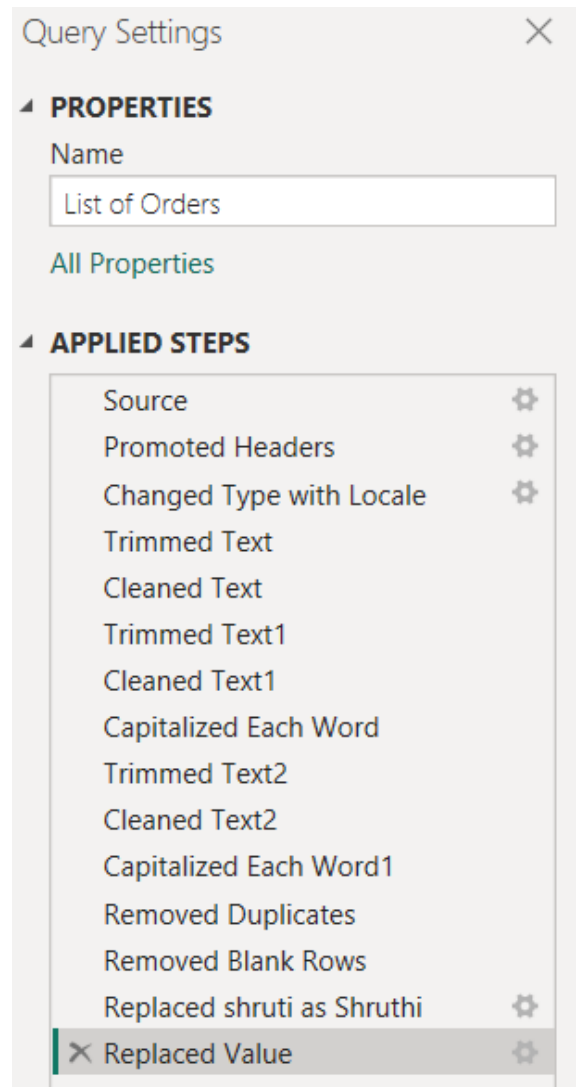


Figure: Power Query applied steps used to clean and transform the source tables

DATA MODEL & RELATIONSHIPS

Fact Tables

- Order Details (Sales transactions)
- Sales Target (Monthly targets)
- Orders (Customer, State, City, Date)

Product Dimension

- Category, Sub-Category, Brand

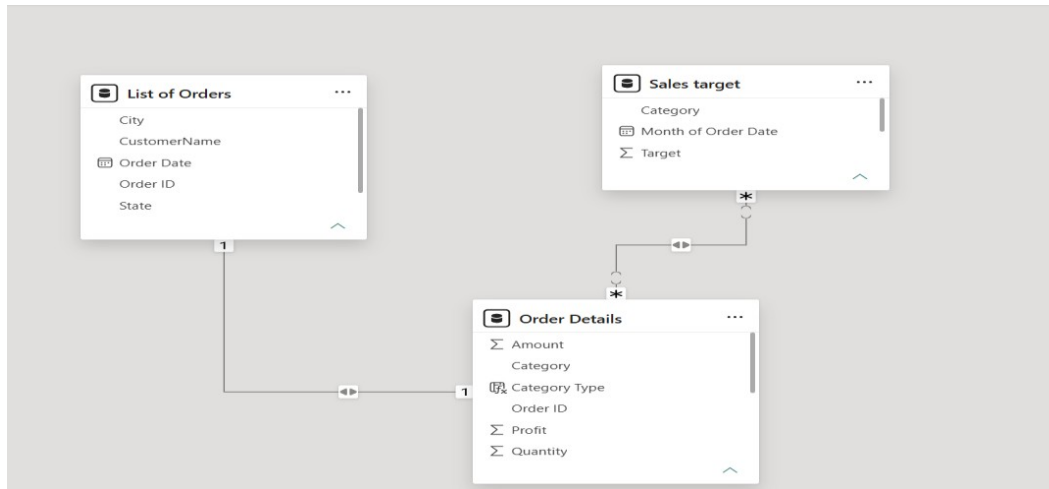


Figure: Data model showing relationships between List of Orders, Order Details, and Sales Target

DAX MEASURES

Calculated Columns

Category Type

```
Category Type = 'Order Details'[Category] & "-" & 'Order Details'[Sub-Category]
```

Revenue Per Order

```
Revenue Per Order = 'Order Details'[Amount] * 'Order Details'[Quantity]
```

Key Measures

Order Count

```
Order Count = COUNT('Order Details'[Order ID])
```

Average Profit in Delhi

```
Average Profit in Delhi = CALCULATE(AVERAGE('Order Details'[Profit]), 'List of Orders'[City]="Delhi")
```

Year-to-Date (YTD) Sales

```
Year To Date Sales = TOTALYTD(SUM('Order Details'[Amount]), 'List of Orders'[Order Date])
```

Total Sales

```
Total Sales = SUM('Order Details'[Amount])
```

Total Target

```
Total Target = SUM('Sales Target'[Target])
```

Target Achievement %

```
Target Achievement % = DIVIDE([Total Sales], SUM('Sales Target'[Target]))
```

YoY Growth %

```
YoY Growth % = DIVIDE([Total Sales] - [Prior Year Sales], [Prior Year Sales])
```

KEY PERFORMANCE INDICATORS (KPIs)

Order Count	500
Total Sales	₹ 1,35,165
Total Target	₹ 1,83,900
Year To Date Sales	₹ 21,624
Average Profit in Delhi	₹ 43.76
Minimum Target	₹ 9,000
Sum of Revenue Per Order	₹ 6,41,035

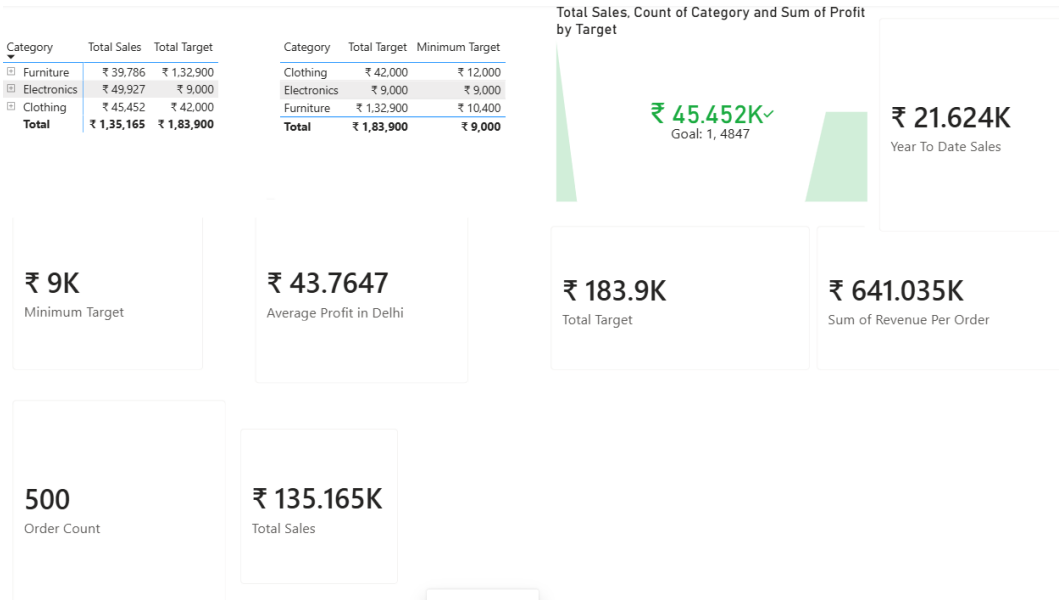


Figure: KPI cards from the Power BI dashboard

SALES BY CATEGORY

Category	Total Sales	Total Target
Clothing	₹ 45,452	₹ 42,000
Electronics	₹ 49,927	₹ 9,000
Furniture	₹ 39,786	₹ 1,32,900
Total	₹ 1,35,165	₹ 1,83,900

Category	Total Sales	Total Target
⊕ Clothing	₹ 45,452	₹ 42,000
⊕ Electronics	₹ 49,927	₹ 9,000
⊕ Furniture	₹ 39,786	₹ 1,32,900
Total	₹ 1,35,165	₹ 1,83,900

Figure: Sales by Category table from the Power BI dashboard

DASHBOARD FEATURES

Page 1: Overview

- Sales Dashboard with high-level KPI cards
- Sales Target comparison table by category

Page 2: Deep Dive

- KPI cards, matrix, and multi-card visuals
- Category-wise breakdown of performance

Pages 3 & 4: Performance

- Profit margin analysis and trend charts
- Geographic and sub-category visualizations

VISUALIZATIONS CREATED

Sales Target Achievement by Category

- Horizontal bar chart comparing Total Sales vs Total Target for Furniture, Clothing, and Electronics
- Furniture shows the highest sales and target values, indicating strong revenue generation
- Clothing performs moderately; Electronics records the lowest figures

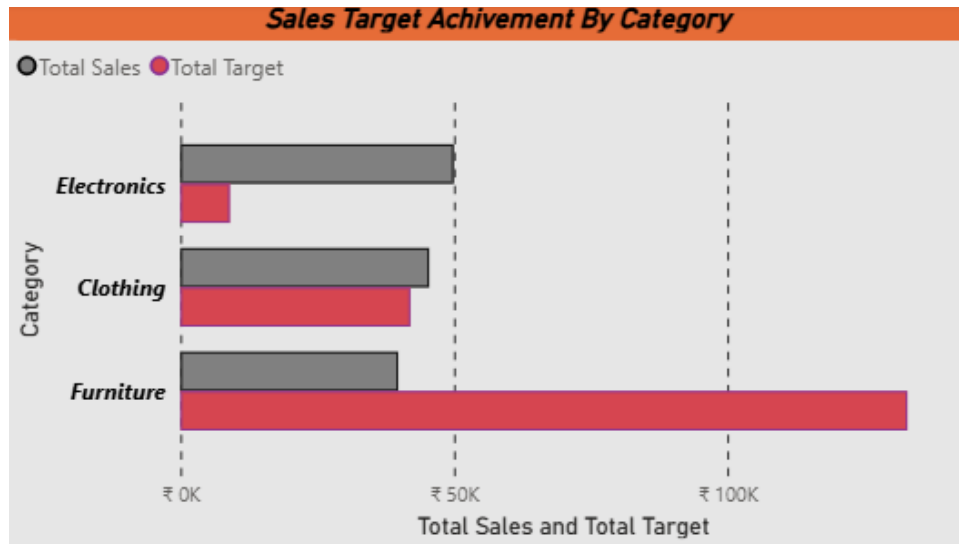


Figure: Sales Target Achievement by Category

Monthly Sales Trends

- Line graph displaying Average of Amount from January 2026 to November 2026 (updated: June 28, 2026)
- Trend shows periodic fluctuations — peaks around ₹450K and dips near ₹200K
- Variations may reflect seasonal demand or promotional cycles

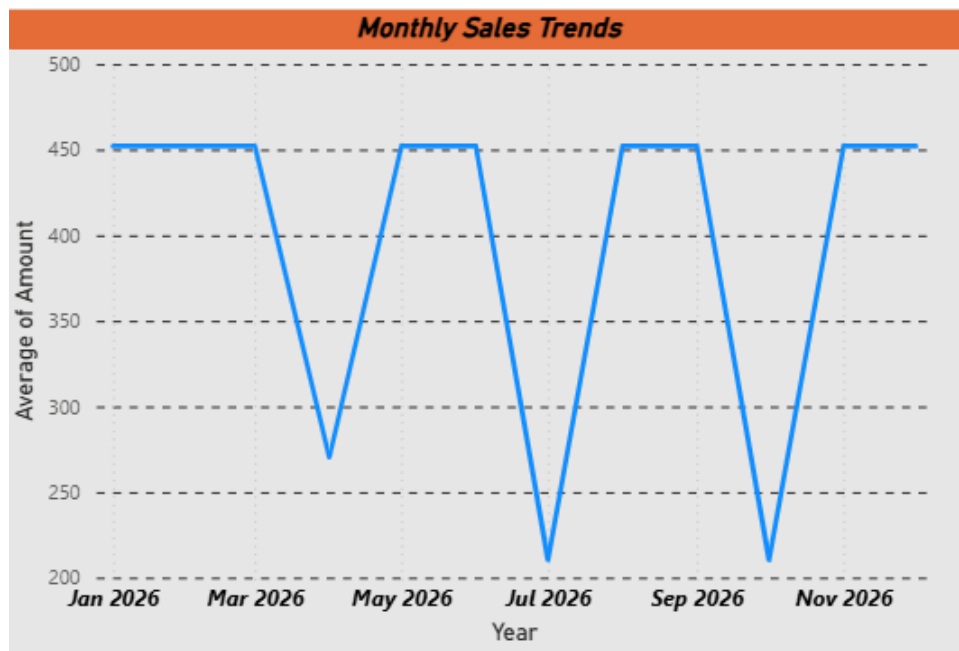


Figure: Monthly Sales Trends

Max Profit Margin by Sub-Category

- Donut chart showing maximum profit margins per sub-category
- Top contributors: T-Shirt (15.04%), Accessories (14.1%), Shirt (10.8%)
- Bookcases and Furnishings have lower margins

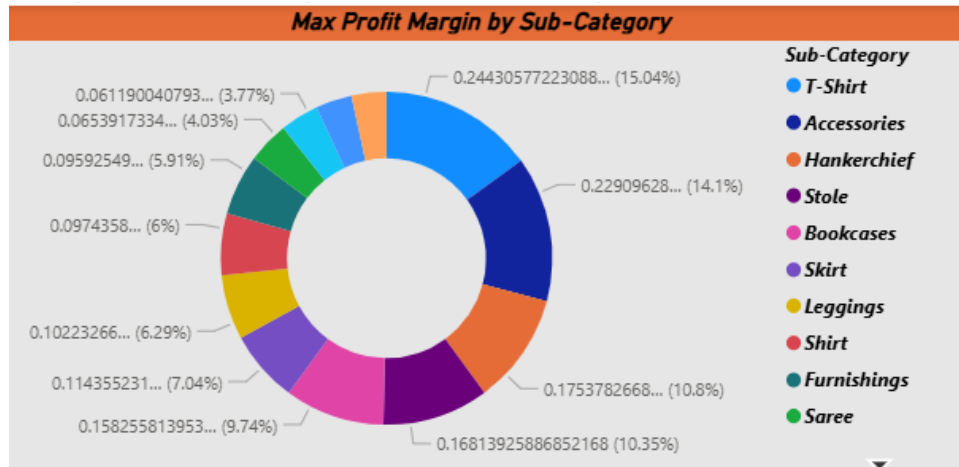


Figure: Max Profit Margin by Sub-Category

Profit vs Quantity by Sub-Category

- Scatter plot comparing Sum of Profit and Sum of Quantity across sub-categories
- Accessories and Bookcases show high profit; Handkerchiefs and Leggings show lower profitability

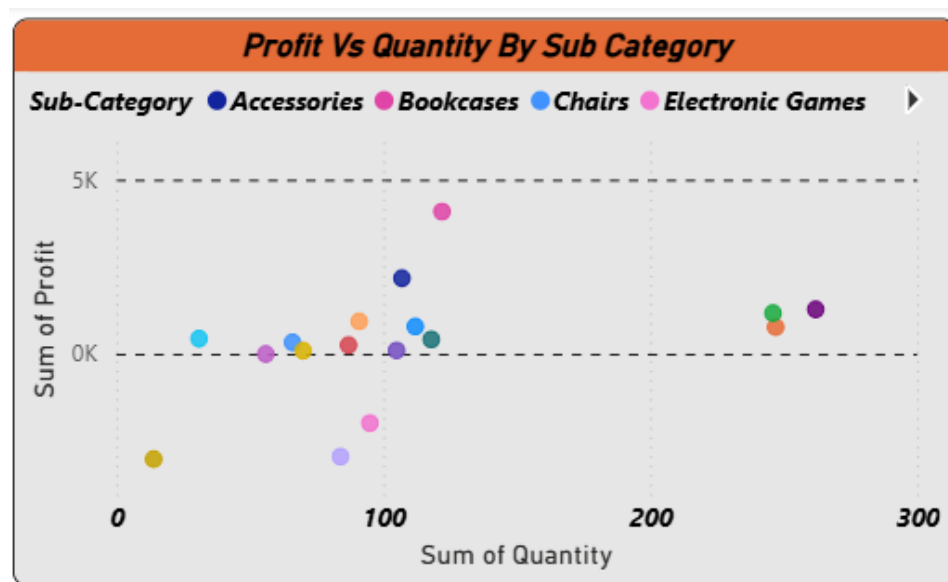


Figure: Profit vs Quantity by Sub-Category

Gauge Chart – Overall Sales Performance

- Displays total sales of ₹641.04K out of ₹1.28M target
- Sales have achieved nearly half the expected target, with room for growth

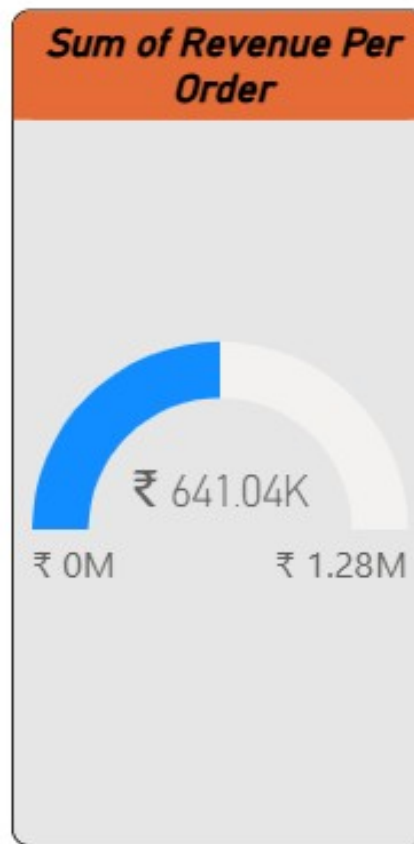


Figure: Gauge Chart – Overall Sales Performance

Geographic Sales Analysis

- Map visual showing Average of Amount by City
- Sales concentrated in Madhya Pradesh, Maharashtra, and Rajasthan



Figure: Geographic Sales Analysis – Average of Amount by City

Sales Distribution by Sub-Category

- Treemap showing relative sales volumes across sub-categories
- Bookcases, Saree, Printers, and Accessories among top contributors

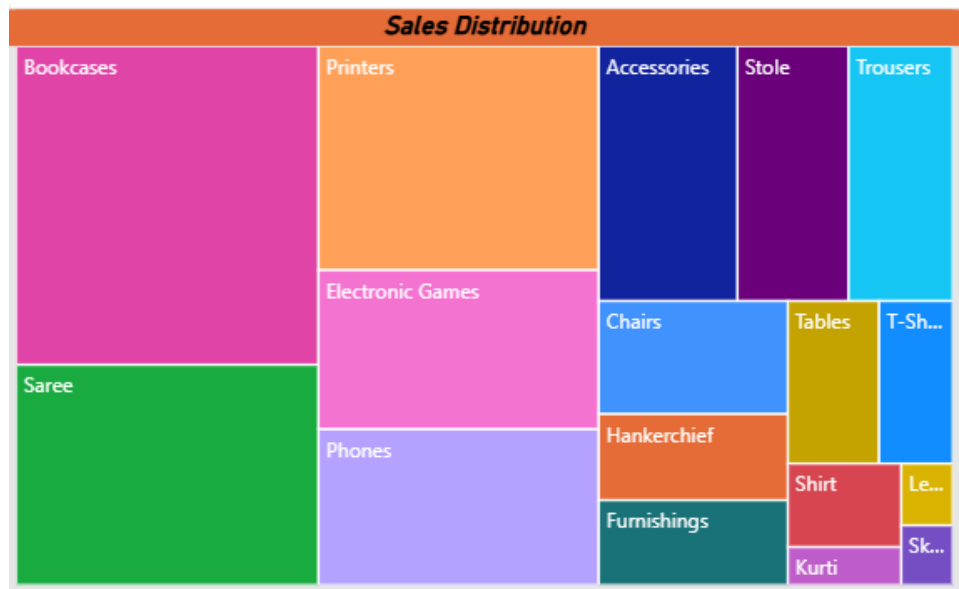


Figure: Sales Distribution by Sub-Category

Order Count by State

- Madhya Pradesh leads with 55 orders — top-performing state
- Maharashtra (19 orders) and Punjab (16 orders) follow
- States like Karnataka, Uttar Pradesh, Gujarat, and Rajasthan: 10–15 orders each
- Delhi, Kerala, Andhra Pradesh, Bihar, Tamil Nadu show opportunities for growth

KEY INSIGHTS

Category Performance

- Furniture achieved the highest total sales but fell short of monthly targets in several periods
- Clothing maintained consistent growth and frequently met or exceeded targets
- Electronics showed fluctuating performance with peaks during promotional months

Profitability Trends

- Clothing contributed most to overall profit; Furniture had high revenue but lower margins
- Printers and Bookcases: high sales but moderate profits — potential pricing adjustments needed

Regional Insights

- Major sales concentrations in Maharashtra, Madhya Pradesh, and Rajasthan

- Cities like Mumbai, Indore, and Jaipur consistently ranked among top performers

Customer Behavior

- Repeat orders and high-value transactions more frequent in metropolitan regions
- Seasonal spikes in April and December suggest festive or promotional influences

CONCLUSION

This Power BI project successfully demonstrates how data analysis and visualization can transform raw datasets into meaningful business insights. By integrating List of Orders, Order Details, and Sales Target data, the dashboard provides a clear view of sales performance across categories, regions, and time periods.

The project highlights the power of DAX measures for dynamic calculations, Power Query for efficient data cleaning, and Power BI's rich visualization capabilities for communicating actionable insights to diverse stakeholders including executive leadership, sales operations, and finance teams.